Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
The clients are unable to access the travel agency website. They can't retrieve content from the web server. This is usually generated by a problem with the web server or a misconfiguration of the firewall. The log file generated with Wireshark shows requests and responses used with the TCP and HTTP protocol. HTTP is based on the TCP protocol for the initial phase handshake protocol, which explains the presence of both protocols. Most of the requests are TCP requests which are abruptly interrupted when the web server requires the ACK response from the client. Most of the requests come from the same IP address "198.51.100.14". Considering the foregoing information, the issue is likely caused by an DoS attack, in particular a SYN Flood, where a threat actor sends malicious packets coming from the same source to flood the web server, and make it unable to process requests coming from legit clients.

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1. First, the client sends a TCP SYN request to the server to attempt to establish connection with it. 2. The web server answers that request with a TCP SYN/ACK response, authorizing the connection attempt of the client, and keeps waiting until his TCP ACK confirmation. 3. The client sends a TCP ACK packet, which confirms and establishes the connection with the server. When a malicious actor sends multiple SYN packets, the server is unable to process all of them because it keeps waiting for the TCP ACK confirmation, but none of them are going to be sent. The foregoing generates a flood in the web server, making it overwhelmed and incapable of processing the following requests, including legitimate requests coming from real clients.

The logs reflect the past description, multiple TCP SYN requests flooding the web server, with no TCP ACK packets, usual behavior for a DoS SYN Flood Attack.

The effects in the organization can be catastrophic, from financial (customers are unable to hire the services of the agency) to reputational (customers could feel the agency as unreliable, and decide to use services of other agencies).

For this, the company should shut off the web server, check the firewall they have if it is installed or if not, install a firewall (ideally a NGFW), and set it up to prevent this attack in the future.